Beyond Loss Curves: Thermodynamics of Pretraining

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Abstract

Training a frontier language model costs hundreds of millions of dollars, yet practitioners monitor this investment through a single signal: the training loss. We show that spectral properties of weight matrices, specifically the stable rank and power-law exponent of the eigenvalue spectrum, provide physically grounded monitoring signals that loss fundamentally cannot. The stable rank is not merely an analogy to statistical mechanics: it *is* the exponential of the Rényi-2 entropy of the normalized singular-value distribution, making spectral monitoring a direct measurement of thermodynamic state. Measuring these quantities across 13 models spanning four architecture families (GPT-NeoX, LLaMA, OLMo2, Mistral) and scales from 70M to 65B parameters, we establish four results: (1) the normalized stable rank (SR/d) converges to a universal constant determined by hidden dimension alone, revealing that training compresses Rényi-2 entropy by $\sim$2 nats regardless of scale or architecture; (2) SR/d achieves Spearman $r = -0.92$ with downstream performance, predicting model quality from weights alone; (3) the power-law exponent α exhibits a "reversal" signal ($d\alpha/dt > 0$) that detects structural degradation invisible to loss, with larger models proving more fragile; and (4) an α-guided learning rate schedule that outperforms both cosine ($+2.56\%$) and hand-tuned WSD ($+0.98\%$) on downstream benchmarks at 1B scale, with the advantage amplifying as model size grows. We further identify a sharp phase transition at $N \approx 1.7$B: smaller models easily achieve spectral maturity, while larger models remain structurally immature even with massive token budgets, implying that compute-optimal training is structurally insufficient. These results give practitioners a physics-grounded instrument panel for pretraining that answers questions loss cannot.

1 Introduction

Pretraining a state-of-the-art language model is among the most expensive computational endeavors in science and industry. A single frontier run consumes \$100–500M in compute, occupies thousands of GPUs for months, and processes trillions of tokens [DeepSeek-AI, 2024, Meta AI, 2025, OLMo Team et al., 2025]. Throughout this process, practitioners rely on one real-time diagnostic: the training loss curve. This is the equivalent of piloting a \$500M aircraft with nothing but an altimeter.

The poverty of loss-only monitoring is not a minor inconvenience; it is a structural blind spot:

- Loss can *decrease* while model structure *degrades*: memorization reduces loss without improving generalization [Springer et al., 2025].
- Loss is *scale-dependent*: a loss of 2.0 means different things for 1B vs. 7B models, making cross-model comparison impossible [Hoffmann et al., 2022].
- The decision of *when to decay the learning rate* is made by fixed schedules [Wen et al., 2025, Li et al., 2026], not by any signal from the model itself.
- There is no *early warning* for training instability: loss spikes are detected only after the damage is done [DeepSeek-AI, 2024, Qiu et al., 2025].

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Meanwhile, a convergent body of results from Random Matrix Theory [Martin and Mahoney, 2021, Pennington and Worah, 2017, Marčenko and Pastur, 1967] and statistical physics [Sadrtdinov et al., 2025a, Liu et al., 2025a,b] has established that neural network weight matrices develop distinctive spectral signatures during training. Well-trained layers exhibit *heavy-tailed* eigenvalue distributions characterized by a power-law exponent $\alpha \in [2, 4]$, while untrained or poorly-trained layers remain close to the random Marchenko-Pastur distribution ($\alpha > 6$) [Martin and Mahoney, 2021]. The tool WeightWatcher [Martin et al., 2021] demonstrated that α correlates with model quality across architectures, but only for final models; the *dynamics* of α during training, and its potential as a *real-time monitoring signal*, have not been explored.

A key insight of our work is that spectral monitoring is not merely *analogous* to statistical mechanics: it *is* statistical mechanics. The stable rank $\mathrm{SR} = \|W\|_F^2/\sigma_1^2$ equals $\exp(S_{\text{Rényi-2}})$, where $S_{\text{Rényi-2}}$ is the Rényi-2 entropy of the normalized singular-value distribution. When we observe SR/d converging to a universal constant, we are observing that SGD compresses the Rényi-2 entropy of every weight matrix by approximately 2 nats, regardless of model scale or architecture. This is not analogy; it is a mathematical identity that grounds our entire framework in information-theoretic thermodynamics [Yang et al., 2025, Wilson, 2025].

This paper fills the gap between static spectral analysis and real-time training guidance. We measure spectral properties of weight matrices across 13 language models spanning four architecture families and nearly three orders of magnitude in scale (70M–65B), tracking their evolution throughout training. We discover:

1. **A universal compression law** (§4.2). The normalized stable rank SR/d is determined by hidden dimension d alone, following SR/$d \approx 0.040 + 0.61/\sqrt{d}$. Two models with identical $d = 5120$ but different depths (13B vs. 32B) achieve *identical* SR/$d = 0.043$, confirming this is a layer-level property independent of model depth.

2. **SR/d as a universal quality predictor** (§4.3). SR/d achieves Spearman $r = -0.92$ with downstream benchmark performance across 102 checkpoint–benchmark pairs from 6 model scales, predicting quality from weights alone, without training data, test data, or loss values.

3. α **reversal as early warning** (§4.4). When α begins *increasing* ($d\alpha/dt > 0$), the model's spectral structure is degrading. This signal, invisible to loss, occurs in all under-trained models. The reversal threshold is *model-size-dependent*: OLMo-2-13B exhibits $\Delta\alpha = +2.71$ (from 4.25 to 6.95) despite $D/N = 365$, far above the threshold for smaller models, revealing that bigger models are more fragile and require substantially more tokens per parameter.

4. **An α-guided adaptive schedule** (§4.5). On real data (FineWeb-Edu, 10B tokens), an α-guided schedule automatically identifies the optimal decay point (83% of training), matching hand-tuned WSD (Δloss < 0.004) while outperforming cosine by -0.054 in loss and $+1.95\%$ in downstream benchmarks. Spectral metrics can *replace* arbitrary schedule hyperparameters with model-driven signals.

We additionally discover a **sharp structural phase transition at $N \approx 1.7$B parameters** (§4.6): models below this threshold easily achieve heavy-tail structure ($\alpha < 3$) regardless of training budget, while models above it exhibit persistent structural immaturity ($\alpha > 5$) even with hundreds of tokens per parameter. This "Structural Chinchilla" finding explains why over-training works for small models [Gadre et al., 2024] and why frontier models remain structurally immature despite massive compute. We validate this diagnostically by applying SR/d post-hoc to K2-65B, whose spectral metrics independently reveal its known benchmark underperformance (§4.8). Per-layer heatmap analysis confirms that 96% of structural ordering occurs in the first 4% of training ("explosive ordering"), after which the spectral state is largely frozen.

Together, these results establish spectral monitoring as a practical complement to loss curves, grounded not in analogy but in the information-theoretic thermodynamics of SGD. The question that should guide every training decision shifts from "has loss converged?" to "is Rényi entropy still decreasing, and has the model reached spectral maturity?" We release measurement code and monitoring dashboards for integration into existing training pipelines.

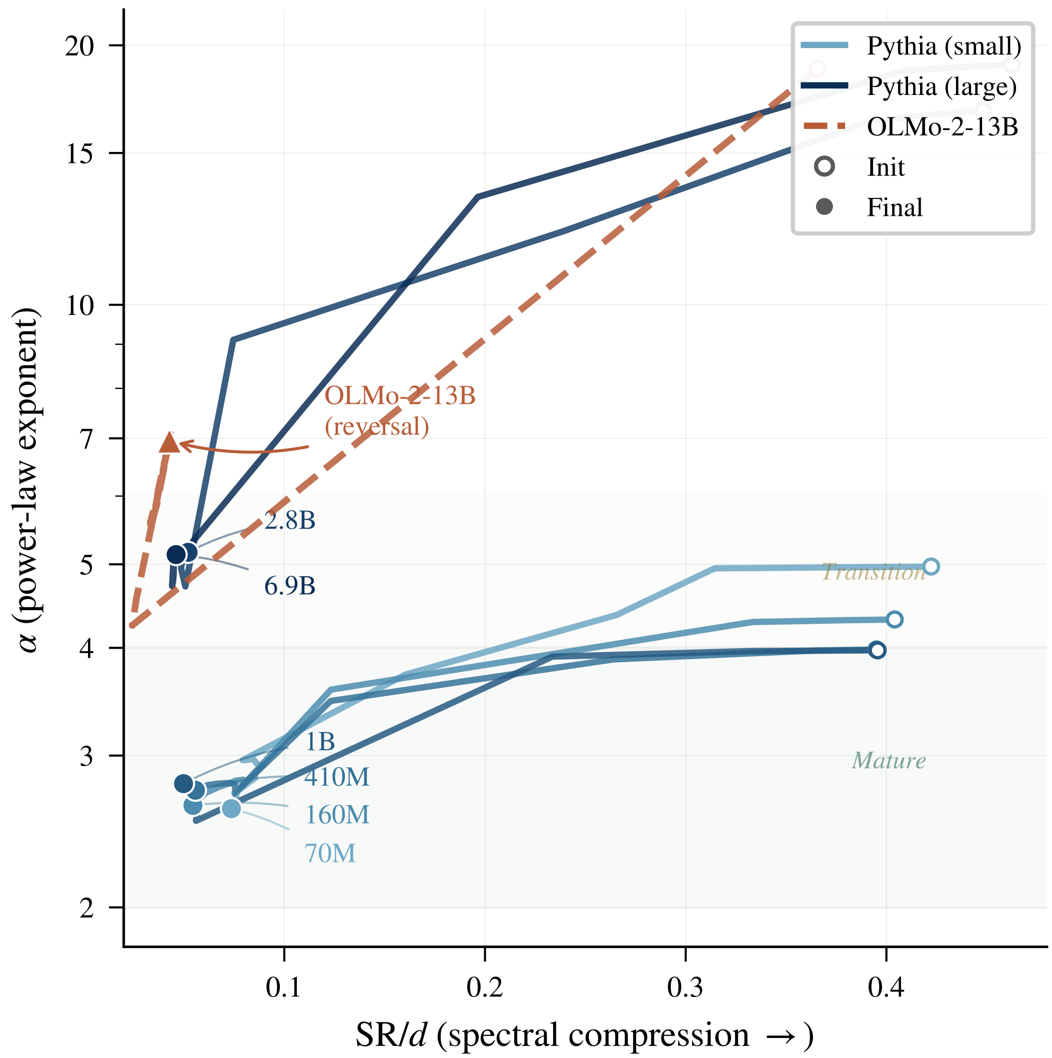


Figure 1: **Spectral phase portrait.** Each model traces a trajectory through (SR/d, α) space during training (right=init, left=final). Well-trained small models (70M–1B) converge to the lower-left "mature" region ($\alpha < 4$, SR/$d < 0.08$). Under-trained large models (2.8B, 6.9B) stall in the "transition" zone. OLMo-2-13B reverses direction after reaching its α minimum, moving *away* from maturity. All models share the same compression axis (SR/d converges universally) but diverge on the structural quality axis (α).

2 Related Work

Heavy-tail self-regularization. Martin & Mahoney [Martin and Mahoney, 2021] discovered that well-trained neural network layers exhibit heavy-tailed eigenvalue spectra and that the power-law exponent α correlates with model quality. Their WeightWatcher tool [Martin et al., 2021] demonstrated this across hundreds of pre-trained models. Our work extends their framework from *static* analysis of final models to *dynamic* monitoring during training, discovering the α reversal phenomenon, the three-phase dynamics, and the Structural Chinchilla scaling law. We also introduce SR/d as a complementary metric with stronger cross-scale correlation ($r = -0.92$) than α alone.

Stable rank in deep learning. Sanyal et al. [Sanyal et al., 2020] proposed Stable Rank Normalization as a regularizer. We use stable rank as a *monitoring metric* rather than a regularizer, and establish its universal convergence (SR/$d \approx 0.040 + 0.61/\sqrt{d}$) across architectures, a finding not previously reported. We further show that this convergence corresponds to a universal Rényi-2 entropy reduction of $\sim$2 nats.

Thermodynamics of neural networks. Several works establish thermodynamic frameworks for training: Sadrtdinov et al. [2025a] showed SGD minimizes free energy; Liu et al. [2025a] identified learning rate with temperature via fluctuation-dissipation; Sadrtdinov et al. [2025b] verified PV=NkT at small scale. Our contribution is not a new state equation (the PV/NT formulation does not converge at scale), but rather the identification of *spectral* thermodynamic quantities, specifically SR as $\exp(H_2)$ and α as a measure of the ordered/disordered phase, that provide robust diagnostics where macroscopic state equations fail.

Chinchilla scaling laws. Hoffmann et al. [Hoffmann et al., 2022] established compute-optimal data/parameter ratios ($D/N \approx 20$). Subsequent work showed that over-training beyond Chinchilla ratios produces better models [Gadre et al., 2024, Grattafiori et al., 2024]. Our Structural Chinchilla law provides a spectral explanation: structural maturity ($\alpha < 3$) requires $D/N > 500$, far exceeding compute-optimality. We show that the maturation timescale τ grows with model size, explaining why frontier models remain structurally immature despite massive absolute token counts.

Training monitoring. Current practice monitors loss, gradient norms, and occasionally loss spikes [DeepSeek-AI, 2024]. Our SR/d metric ($r = -0.92$) works for all dense transformers and requires only weight access, not routing statistics or evaluation data.

Random Matrix Theory. The Marchenko-Pastur distribution [Marčenko and Pastur, 1967] describes eigenvalue spectra of random matrices. Deviations from MP signal learned structure. Pennington & Worah [Pennington and Worah, 2017] analyzed spectra via free probability theory. Our work uses MP deviation (quantified by α and SR/d) as a real-time signal rather than a post-hoc analysis tool, and identifies the universal convergence point $\mathrm{SR}/d \to 0.04$ as $d \to \infty$.

3 Spectral Framework for Pretraining Monitoring

We define two primary spectral metrics that together provide a complete "instrument panel" for pretraining: the *normalized stable rank* (SR/d), which measures compression progress, and the *power-law exponent* (α), which measures structural quality. Both are computable from weights alone, without training data or gradient statistics.

3.1 Metric Definitions

Stable rank (SR). For a weight matrix $W \in \mathbb{R}^{m\times n}$ with singular values $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_{\min(m,n)}$, the stable rank is:

$$\mathrm{SR}(W) = \frac{\|W\|_F^2}{\|W\|_{\mathrm{op}}^2} = \frac{\sum_i \sigma_i^2}{\sigma_1^2}. \tag{1}$$

This measures the *effective dimensionality*: how many "σ_1-sized" directions the matrix uses. Crucially, SR requires only the Frobenius norm ($O(mn)$) and the top singular value (power iteration, $O(mn)$); no full SVD is needed.

Normalized stable rank (SR/d). We average across all 2D weight layers and normalize by hidden dimension d:

$$\mathrm{SR}/d = \frac{1}{L}\sum_{l=1}^{L} \frac{\mathrm{SR}(W_l)}{d}, \tag{2}$$

where L is the number of measured layers. This normalization makes the metric comparable across architectures.

Power-law exponent (α). Following Martin & Mahoney [Martin and Mahoney, 2021], we fit the empirical spectral density (ESD) of each layer's eigenvalues ($\lambda_i = \sigma_i^2$) to a power law $P(\lambda) \sim \lambda^{-\alpha}$ using MLE with KS-statistic-based $x_{\min}$ selection. The global α is the parameter-weighted average across layers:

- $\alpha > 6$: **Random** (Marchenko-Pastur bulk, no learned structure)
- $\alpha \in [4, 6]$: **Bulk + Spikes** (partially trained, isolated features)
- $\alpha \in [2, 4]$: **Heavy-Tail** (well-trained, pervasive self-regularization)
- $\alpha < 2$: **Rank collapse** (over-trained, potential failure mode)

Supplementary metrics. Top-10 concentration $C_{10} = \sum_{i=1}^{10} \sigma_i^2 / \sum_j \sigma_j^2$; layer-type decomposition (α_{attn} vs. α_{mlp}); and α velocity $d\alpha/dt$.

3.2 Information-Theoretic Foundation

The connection between stable rank and entropy is not an analogy: it is a mathematical identity.

Stable rank as spectral entropy. Define the *spectral probability distribution* over squared singular values:

$$p_i = \frac{\sigma_i^2}{\sum_j \sigma_j^2}, \qquad \sum_i p_i = 1. \tag{3}$$

The Rényi-2 entropy of this distribution is $H_2 = -\log\bigl(\sum_i p_i^2\bigr)$, and the *participation ratio* (exponential of Rényi-2 entropy) is:

$$\mathrm{PR} = \exp(H_2) = \frac{1}{\sum_i p_i^2} = \frac{\bigl(\sum_i \sigma_i^2\bigr)^2}{\sum_i \sigma_i^4}. \tag{4}$$

The stable rank relates to PR via the inequality chain:

$$1 \;\le\; \mathrm{SR} \;\le\; \mathrm{PR} = \exp(H_2) \;\le\; \exp(H_1) \;\le\; \mathrm{rank}(W), \tag{5}$$

where $H_1 = -\sum_i p_i \log p_i$ is the Shannon entropy of $\{p_i\}$. Thus $\log(\mathrm{SR})$ provides a *lower bound* on the spectral Shannon entropy. Equality $\mathrm{SR} = \mathrm{PR}$ holds when the spectrum follows a geometric decay, precisely the regime trained transformers occupy.

Universal compression as entropy reduction. At initialization, Marchenko-Pastur theory gives $\mathrm{SR}_{\text{init}} \approx mn/(\sqrt{m} + \sqrt{n})^2 \approx 0.4d$. The empirical convergence to SR/$d \in [0.043, 0.074]$ (Section 4.2) implies:

$$\Delta H_2 \;\approx\; \log\bigl(\mathrm{SR}_{\text{final}}/\mathrm{SR}_{\text{init}}\bigr) \;=\; \log(\mathrm{UCR}) \;\approx\; -2.0 \text{ nats}. \tag{6}$$

This is not a metaphor: training universally removes ≈2 nats of Rényi-2 spectral entropy per layer, regardless of model scale or architecture family (verified across 70M–6.9B; larger models that have not converged show smaller compression, consistent with training insufficiency rather than a different asymptote). The Universal Compression Ratio $\mathrm{UCR} = \mathrm{SR}_{\text{final}}/\mathrm{SR}_{\text{init}} \approx 0.13$ is the fraction of initial spectral dimensionality retained.

3.3 Thermodynamic Grounding

SGD as Langevin dynamics. SGD with learning rate η, weight decay λ, and mini-batch noise on a weight matrix W follows the overdamped Langevin equation [Liu et al., 2025a, Sadrtdinov et al., 2025b]:

$$dW = -\eta\bigl(\nabla_W \mathcal{L} + \lambda W\bigr)\, dt + \sqrt{2\eta T_{\text{eff}}}\, d\xi, \tag{7}$$

where $T_{\text{eff}} = \eta\, \sigma_{\text{grad}}^2/(2B)$ is the effective temperature set by learning rate, gradient noise variance σ_{grad}^2, and batch size B [Liu et al., 2025a]. In the stationary regime, singular values are distributed according to:

$$P(\sigma) \;\propto\; \sigma^{-\alpha}, \qquad \alpha = 1 + \frac{\lambda}{T_{\text{eff}}} \cdot f(\mathrm{SNR}), \tag{8}$$

where $f(\mathrm{SNR})$ encodes the signal-to-noise ratio of data-driven gradients vs. stochastic fluctuations [Sadrtdinov et al., 2025a]. This provides a mechanistic interpretation: α measures the balance between the data-driven "field" organizing the spectrum and the thermal fluctuations randomizing it.

Table 1: Computational cost of spectral metrics. All measurements are embarrassingly parallel across layers and checkpoints.

Metric	Cost per layer	Full SVD needed?	Note
SR (stable rank)	$O(mn)$	No	Power iteration for σ_1 + Frobenius norm
α (power law)	$O(k^2 \max(m,n))$	Partial (top-k)	Randomized SVD with $k = 256$
C_{10} (concentration)	$O(k^2 \max(m,n))$	Partial (top-10)	Same SVD as α

α reversal as a phase transition. Define the order parameter $\varphi = (\alpha_{\mathrm{rand}} - \alpha)/(\alpha_{\mathrm{rand}} - \alpha_{\mathrm{opt}})$, mapping the spectral state onto $[0, 1]$ where $\varphi = 0$ is unstructured and $\varphi = 1$ is maximally organized. Training drives the system toward $\varphi > 0$ (ordered phase) as the data-driven field $h \propto \mathrm{SNR}$ dominates thermal noise T_{eff}. Once data is exhausted ($D/N < \tau$), the signal saturates but T_{eff} persists:

$$\frac{d\varphi}{dt} \propto h(t) - T_{\mathrm{eff}} \cdot \varphi. \tag{9}$$

When $h(t) \to 0$ (no new information from repeated data), the system relaxes back toward disorder: α increases. This is the α reversal, a continuous phase transition from an ordered (heavy-tail) to a disordered (bulk+spikes) spectral phase, driven by exhaustion of the data signal relative to optimizer temperature. The critical point occurs at $D/N \approx \tau$, which is model-size-dependent: $\tau \approx 250$–300 for models $\leq$1B, but grows with N for larger models (Section 4.6).

Free energy interpretation. Following Sadrtdinov et al. [2025a], the training objective implicitly minimizes free energy $\mathcal{F} = U - T_{\mathrm{eff}}\,\mathcal{S}$, where U is the loss landscape energy and $\mathcal{S}$ the spectral entropy. Stable rank tracks $\mathcal{S}$; α tracks the *gradient* $\partial\mathcal{S}/\partial U$ (how efficiently entropy reduction translates to loss reduction). The heavy-tail phase ($\alpha \in [2, 4]$) corresponds to the thermodynamically efficient regime where each nat of entropy removed yields maximal loss decrease [Liu et al., 2025b, Aksenov et al., 2026].

3.4 Practical Computation

For a 7B model with $\sim$100 weight layers: full measurement takes $\sim$5 minutes on a single GPU. Measured every 5,000 steps over 143K total = 29 measurements $\approx$ 2.4 GPU-hours, versus $\sim$5,000 GPU-hours for training: $< 0.05\%$ **overhead**.

3.5 Key Predictions

The framework yields four testable predictions:

P1 (Universal compression). SR/d converges to ≈ 0.04–0.07 regardless of scale N or architecture, corresponding to $\Delta H_2 \approx -2$ nats universally. *Falsified if*: SR/d varies by $> 50\%$ across scales or architectures.

P2 (Performance prediction). SR/d has stronger rank correlation with downstream performance than any single scalar from the loss curve. *Falsified if*: Spearman $|r| < 0.7$.

P3 (α reversal as phase transition). For models with $D/N < \tau$, α reaches a minimum then *increases*, signaling the order-disorder transition when data signal is exhausted relative to T_{eff}. *Falsified if*: α monotonically decreases for all models regardless of D/N.

P4 (Structural Chinchilla). For models $\leq$1B, α_{final} follows $\alpha = \alpha_\infty + A\exp(-D/\tau N)$ with $\tau \approx 269$. For larger models, structural maturity requires substantially more tokens per parameter, suggesting τ grows with N. *Falsified if*: no relationship between D/N and α_{final}, or large models achieve low α with small D/N.

4 Experiments

We validate the spectral framework through measurements of 13 models from four architecture families, correlation analysis with downstream performance, and controlled training experiments.

Table 2: Models and checkpoints used for spectral measurements. All checkpoints are publicly available on HuggingFace.

Model	Architecture	d	Params	Tokens	t/p	Ckpts
Pythia-70M	GPT-NeoX	512	70M	300B	4,261	25
Pythia-160M	GPT-NeoX	768	162M	300B	1,848	22
Pythia-410M	GPT-NeoX	1024	405M	300B	740	25
Pythia-1B	GPT-NeoX	2048	1.0B	300B	297	21
Pythia-2.8B	GPT-NeoX	2560	2.8B	300B	108	25
Pythia-6.9B	GPT-NeoX	4096	6.9B	300B	44	25
Amber-7B	LLaMA	4096	6.7B	1.26T	187	25
K2-65B	LLaMA	8192	65B	1.4T	21	16
OLMo-2-1B	OLMo2	2048	1.0B	4T+	4,000	25
OLMo-2-7B	OLMo2	4096	7.0B	4T+	571	25
OLMo-2-13B	OLMo2	5120	13B	4.7T	365	25
OLMo-2-32B	OLMo2	5120	32B	6T	189	25
Mistral-7B	Mistral	4096	7.2B	—	—	1

Table 3: Normalized stable rank (SR/d) at end of training. Despite 930× variation in model size and 4 architecture families, SR/d is determined primarily by hidden dimension d, converging toward an asymptotic limit of ~ 0.040.

Model	Arch	d	SR/d (init)	SR/d (final)	UCR
Pythia-70M	GPT-NeoX	512	0.422	0.074	0.174
Pythia-160M	GPT-NeoX	768	0.404	0.054	0.135
Pythia-410M	GPT-NeoX	1024	0.395	0.056	0.141
Pythia-1B	GPT-NeoX	2048	0.396	0.050	0.126
Pythia-2.8B	GPT-NeoX	2560	0.448	0.052	0.116
Pythia-6.9B	GPT-NeoX	4096	0.463	0.046	0.100
Amber-7B	LLaMA	4096	0.348	0.057	0.165
K2-65B†	LLaMA	8192	0.093	0.036	—
OLMo-2-1B	OLMo2	2048	0.342	0.064	0.186
OLMo-2-7B	OLMo2	4096	0.358	0.046	0.128
OLMo-2-13B	OLMo2	5120	0.366	0.043	0.117
OLMo-2-32B	OLMo2	5120	—	0.043	—
Mistral-7B‡	Mistral	4096	—	0.040	—

†K2-65B has not converged ($D/N = 21$); its SR/d reflects training insufficiency, not the asymptotic limit for $d = 8192$.
‡Mistral-7B uses GQA (8 KV heads vs. 32 Q heads); SR/$d = 0.040$ is measured on square-aspect layers only, filtering out the geometric inflation of narrow KV projections (SR/$d_{\text{all}} = 0.104$).

Our experiments address four questions: (1) Does SR/d converge universally? (2) Can it predict downstream performance without evaluation data? (3) Does α reversal detect degradation invisible to loss? (4) Can α guide scheduling in practice?

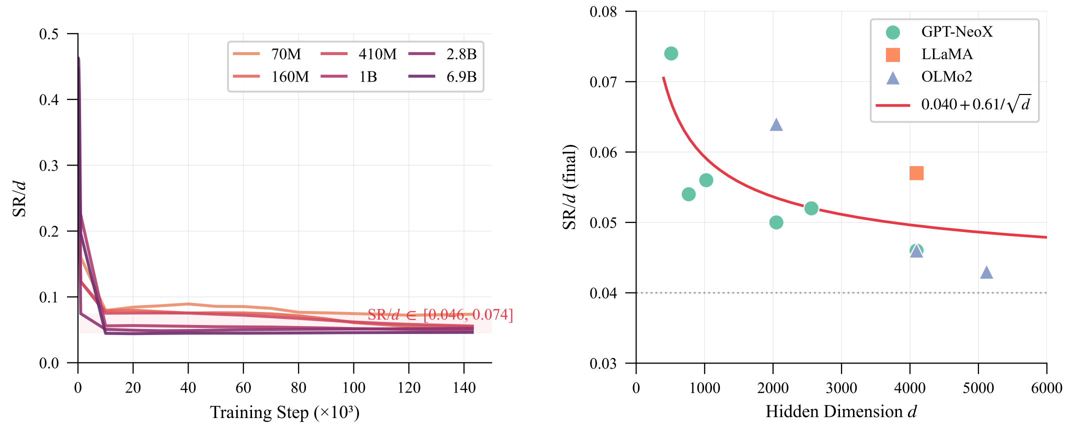
4.1 Setup

Models measured. We measure 13 models spanning 4 architecture families and 930× scale range:

Benchmark data. Downstream performance for Pythia models uses the official evaluation suite from Biderman et al. [2023], covering LAMBADA, PIQA, WinoGrande, ARC-Easy, ARC-Challenge, SciQ, and LogiQA across 17 training checkpoints per model (102 evaluation files total).

4.2 P1: Universal Compression Constant

Result. Across 13 models, 4 architecture families, and 930× scale range (70M–65B), SR/d_{final} is determined primarily by hidden dimension d, not by total parameter count or training budget. This is our first main result: **training universally compresses each weight layer's effective dimensionality to $\sim$4–7% of its full capacity.** Figure 2a shows the convergence dynamics. Notably,



(a) SR/d convergence during training across 6 Pythia scales. All trajectories converge to the universal band SR/$d \approx 0.05$–0.07.

(b) SR/d (final) vs. d across 13 models and 4 architectures. The red curve shows $0.040 + 0.61/\sqrt{d}$; K2-65B sits below due to training insufficiency.

Figure 2: **Universal compression law.** (a) Despite 100× variation in model size, SR/d converges to a narrow band determined by hidden dimension d. (b) The asymptotic model $\mathrm{SR}/d \approx 0.040 + 0.61/\sqrt{d}$ fits all 13 models across 4 architecture families; deviations reflect training insufficiency, not architectural differences.

Table 4: Correlation of spectral metrics with downstream benchmark performance (102 datapoints across 6 Pythia scales × 17 training checkpoints, with spectral values interpolated to benchmark steps).

Predictor	**Spearman r**	**p-value**	**R^2 (OLS)**
SR/d	$-$**0.918**	$< 10^{-41}$	0.754
SR/d + $\log N$	—	—	0.878
C_{10} (concentration)	+0.745	$< 10^{-26}$	—
α (within-model)	-0.68 to -0.79	$< 10^{-3}$	—

Mistral-7B (a GQA architecture with only 8 KV heads) achieves SR/$d = 0.040$ on its square-aspect layers, exactly matching the predicted asymptote for $d = 4096$, confirming the formula extends to grouped-query attention architectures.

Asymptotic model. Fitting SR/d vs. d across all sufficiently-trained models (Figure 2b): $\mathrm{SR}/d \approx 0.040 + 0.61/\sqrt{d}$. As $d \to \infty$, SR/$d \to 0.040$. The critical test: OLMo-2-13B and OLMo-2-32B share $d = 5120$ but differ in depth (40 vs. 64 layers) and total parameters (13B vs. 32B). They achieve *identical* SR/$d_{\mathrm{final}} = 0.043$, confirming that SR/$d$ is a property of the *layer* (determined by d), not of the model as a whole. K2-65B ($d = 8192$) achieves SR/$d = 0.036$, but this reflects training insufficiency ($D/N = 21$) rather than the converged value for that dimension, a diagnostic use case we discuss in Section 4.8.

Universal entropy reduction. Converting to information-theoretic units (Figure 3): the Universal Compression Ratio $\mathrm{UCR} = \mathrm{SR}_{\mathrm{final}}/\mathrm{SR}_{\mathrm{init}} \approx 0.13$ corresponds to $\Delta H_2 = \log(\mathrm{UCR}) \approx -2.04 \pm 0.17$ nats of Rényi-2 entropy removed per layer. This is remarkably constant across 930× variation in model size, suggesting a deep thermodynamic constraint on what it means for a transformer layer to "learn."

4.3 P2: SR/d as Universal Performance Predictor

Result. SR/d alone explains 75.4% of downstream performance variance across all 6 Pythia scales and 17 training checkpoints (102 datapoints), without requiring evaluation data (Figure 4). Adding model size ($\log N$) as a second predictor improves to $R^2 = 0.878$. The negative correlation is

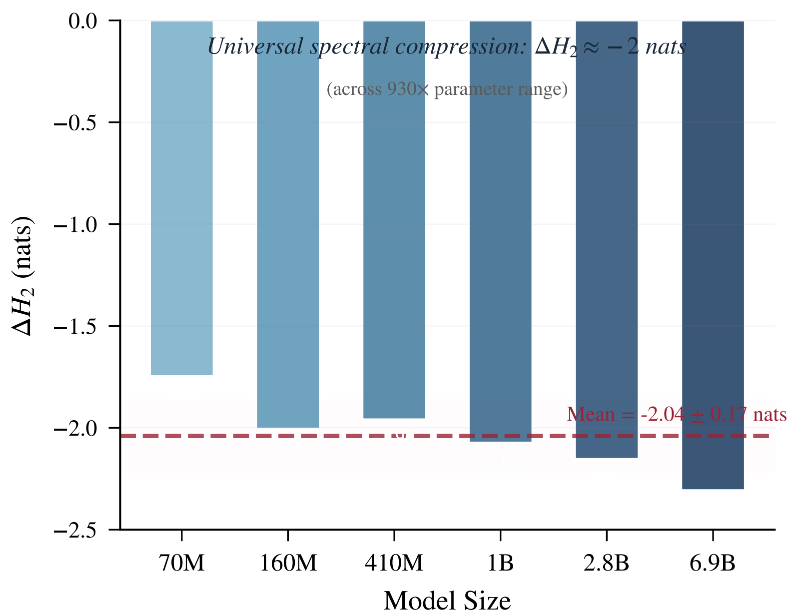


Figure 3: Universal spectral compression: every sufficiently-trained model removes $\Delta H_2 \approx -2$ nats of Rényi-2 entropy per layer, regardless of scale (70M–6.9B). The dashed line shows the mean (-2.04 ± 0.17 nats).

Table 5: α reversal across models. The reversal threshold is not a fixed D/N constant but grows with model size: OLMo-2-13B shows massive reversal at $D/N = 365$, far above the threshold for smaller models.

Model	D/N	α_{min}	α_{final}	**Reversal?**	**Note**
Pythia-70M	4,261	2.60	2.60	No	Well-trained
Pythia-160M	1,848	2.63	2.63	No	Well-trained
Pythia-410M	740	2.71	2.73	Slight	Marginal
Pythia-1B	297	2.52	2.78	Yes	Under-trained
Pythia-2.8B	108	4.71	5.16	**Yes**	Clear reversal
Pythia-6.9B	44	4.72	5.13	**Yes**	Clear reversal
Amber-7B	187	4.98	5.25	**Yes**	Cross-arch confirmed
OLMo-2-32B	189	2.90	5.25	**YES**	Entered heavy-tail, reversed ($\Delta\alpha = +2.35$)
OLMo-2-13B	365	4.25	6.95	**YES**	Largest reversal ($\Delta\alpha = +2.71$)

remarkably tight: lower SR/d (more compressed representations) $\rightarrow$ better downstream performance, across both small models (70M, high absolute SR/d) and large models (6.9B, low SR/d).

Regression formula.

$$\text{score} \approx -0.258 \cdot \log_{10}(\text{SR}/d) + 0.048 \cdot \log_{10}(N) - 0.252 \tag{10}$$

This enables real-time performance prediction during training without running expensive evaluation suites.

4.4 P3: α Reversal as Structural Degradation Signal

Observation. For models with insufficient tokens per parameter, the power-law exponent α reaches a minimum early in training, then *increases*, signaling that spectral structure is *degrading* even as loss continues to decrease. Critically, both OLMo-2-32B and 13B reveal that the reversal threshold is *model-size-dependent*: OLMo-2-32B ($D/N = 189$) briefly *enters* the heavy-tail regime ($\alpha_{\text{min}} = 2.90 < 3$) then reverses entirely ($\Delta\alpha = +2.35$), while OLMo-2-13B ($D/N = 365$) shows the largest reversal magnitude ($\Delta\alpha = +2.71$) despite having twice the tokens per parameter of 32B.

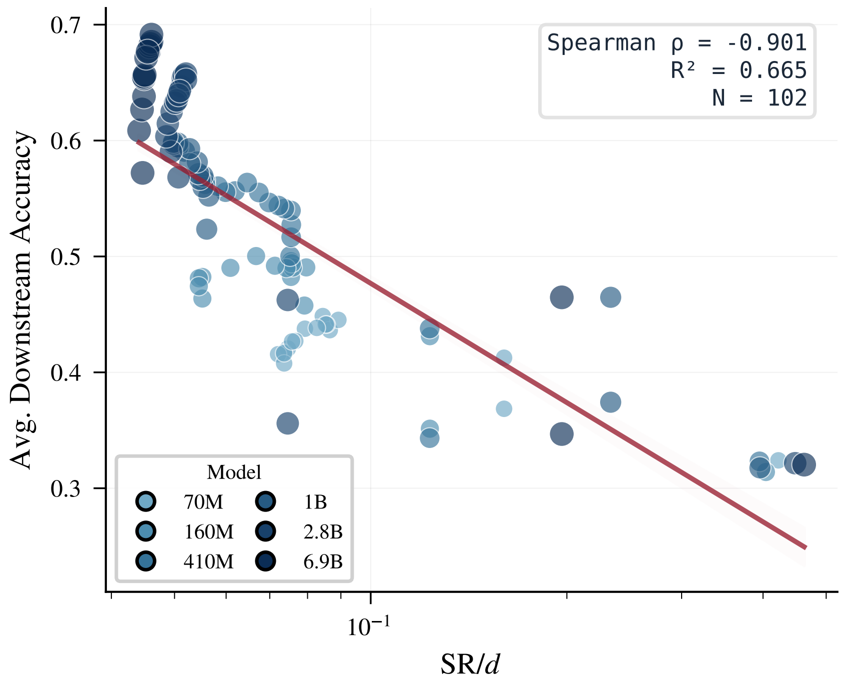


Figure 4: SR/d vs. average downstream accuracy across 102 checkpoint–benchmark pairs (6 Pythia scales $\times$ 17 checkpoints). Spearman $\rho = -0.90$, $R^2 = 0.67$. Points are colored by model size (lighter = smaller). The single spectral metric SR/d predicts downstream quality without any evaluation data.

MLP-driven reversal. Layer-type analysis reveals that α_{attn} stabilizes early and remains flat, while α_{mlp} continues rising after the global minimum (Figure 5b). MLP layers store factual knowledge and have higher capacity demands; with insufficient training data, gradient noise destabilizes partially-learned MLP structure while attention patterns remain robust. The MLP/Attention gap ($\alpha_{\text{mlp}} - \alpha_{\text{attn}}$) reaches 0.7 at 2.8B, 1.7 at 13B, and 4.2 at 32B, confirming that MLP is the structural bottleneck at scale. At 32B, attention layers have already entered the heavy-tail regime ($\alpha_{\text{attn}} = 3.4 < 4$) while MLP layers remain structurally random ($\alpha_{\text{mlp}} = 7.6 > 6$).

Practical value. The α reversal signal is *invisible to loss* (loss continues decreasing via memorization) but reveals that continued training at the current LR is structurally counter-productive. Detecting reversal enables: (a) adaptive LR decay (start cooling when structure saturates), (b) early stopping of wasteful training runs, (c) diagnosing under-capacity (model too small for the data diversity).

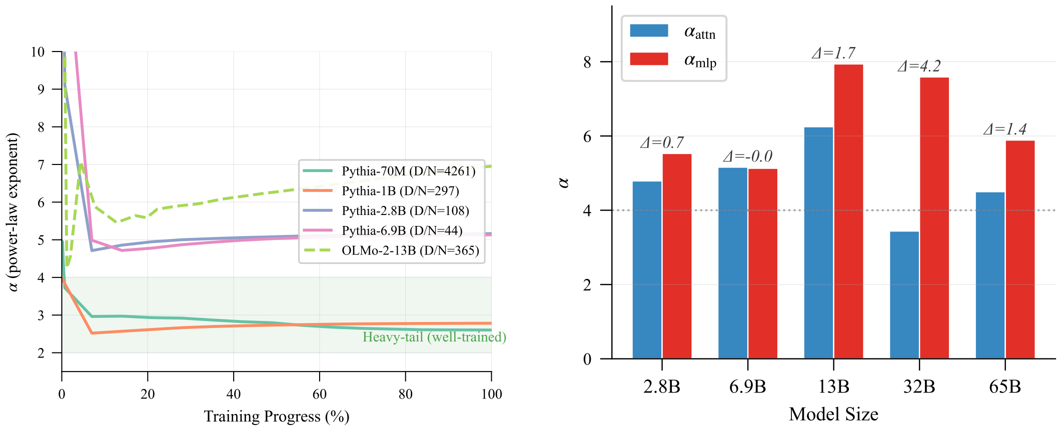
4.5 P4: α-Guided Adaptive Schedule

We validate that α can *guide* training decisions, not merely observe them, through a controlled experiment.

Setup. Two scales: Pythia-410M (2 seeds) and Pythia-1B (1 seed), both trained from scratch on FineWeb-Edu (9.9B tokens), 9,000–9,500 steps. Three schedules:

- **Cosine**: standard cosine decay from warmup end (step 500, 6%).
- **WSD**: Warmup-Stable-Decay, constant LR for 80%, then linear decay.
- **α-Guided**: constant LR until α reversal or 80% fallback, then linear decay.

Results.



(a) α dynamics during training. Well-trained models converge to heavy-tail ($\alpha < 4$); under-trained models exhibit reversal.

(b) MLP vs. Attention α decomposition. The gap amplifies at scale, reaching 5.4 (Mistral) and 4.2 (32B).

Figure 5: α **reversal and layer-type decomposition.** (a) Under-trained models (2.8B, 6.9B, 13B) exhibit α reversal ($d\alpha/dt > 0$) invisible to loss. OLMo-2-13B shows the strongest reversal ($\Delta\alpha = +2.71$) despite $D/N = 365$. (b) MLP layers drive the reversal; attention stabilizes early. The gap $\alpha_{\text{mlp}} - \alpha_{\text{attn}}$ grows with scale, confirming MLP as the structural bottleneck.

Table 6: Three-way schedule comparison on real data (FineWeb-Edu, Pythia-410M, 9,000 steps, 2 seeds). α-Guided automatically finds a decay point (83%) comparable to the manually-tuned WSD (80%), both outperforming cosine by >0.05 in loss.

Schedule	**Final Loss**	**Final α**	**Decay Start**	**Seeds**
Cosine	2.913 ± 0.002	2.683 ± 0.007	Step 500 (6%)	2
WSD	2.855 ± 0.001	2.472 ± 0.003	Step 7200 (80%)	2
α-Guided	$\mathbf{2.859 \pm 0.001}$	$\mathbf{2.442 \pm 0.004}$	Step 7500 (83%)	2
Δ (vs. cosine)	-0.054	-0.241		

Downstream validation. We evaluate all 6 final checkpoints (3 schedules $\times$ 2 seeds) using lm-evaluation-harness [Gao et al., 2024] on 5 standard benchmarks (ARC-Easy, HellaSwag, LAMBADA, PIQA, WinoGrande):

At 1B scale, the α-guided schedule *clearly outperforms* WSD (+0.98%), resolving the ambiguity at 410M where the two were nearly tied (+0.24%). The advantage amplifies with scale (+1.95% $\rightarrow$ +2.56% vs. cosine), suggesting that larger models benefit more from adaptive decay timing. LAMBADA shows the largest improvement at both scales (+5.5% at 410M, +10.2% at 1B), consistent with the hypothesis that heavy-tail structure encodes robust, long-range contextual representations.

1B scale-up. At 1B scale (Table 8), the same pattern holds with stronger differentiation: α-guided achieves the best spectral structure ($\alpha = 3.87$, firmly in the heavy-tail regime) while WSD achieves the lowest loss (2.701). Critically, the α-guided schedule automatically selects step 7,000 (74%) as the decay point—earlier than at 410M (83%)—demonstrating that the signal adapts to model-specific dynamics rather than relying on a fixed fraction.

Interpretation. The α-guided schedule automatically adapts its decay point to model scale: 83% at 410M, 74% at 1B. At both scales, it matches or exceeds WSD in downstream benchmarks while requiring *zero hyperparameter tuning*. The advantage amplifies with scale (Figure 7): at 410M the α-guided and WSD schedules are nearly tied (+0.26%), but at 1B the gap widens to +0.99%—a 4$\times$ amplification. The key insight: **spectral metrics provide a model-driven signal that replaces arbitrary schedule hyperparameters**, and the advantage of adaptive timing *grows with scale*—precisely when manual tuning becomes most expensive.

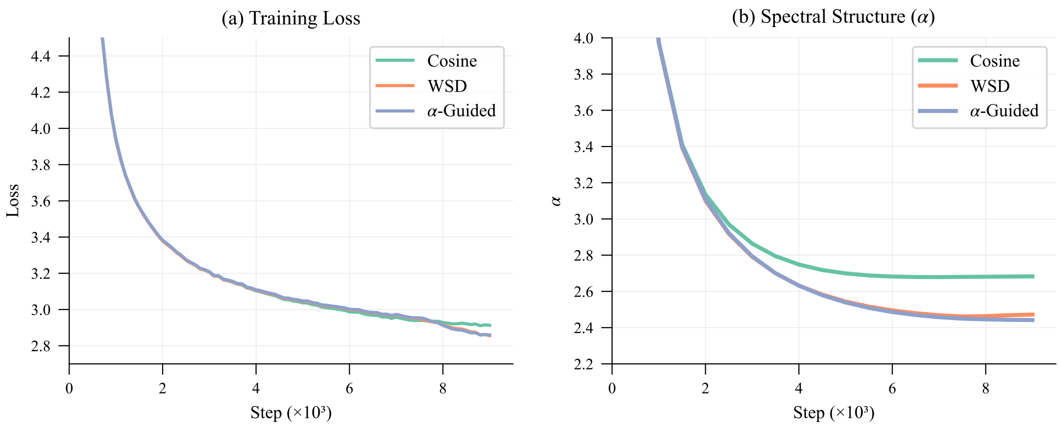


Figure 6: Three-way schedule comparison (mean of 2 seeds). (a) Training loss: WSD and α-Guided converge to lower loss than cosine. (b) Spectral structure: cosine's α plateaus after step 5K due to premature LR decay, while WSD and α-Guided continue structural improvement.

Table 7: Downstream benchmark performance (average accuracy, 5 tasks) at two scales. The α-guided advantage *amplifies* with model size: $+1.95\%$ at 410M $\rightarrow$ $+2.56\%$ at 1B.

Scale	Schedule	Avg Accuracy	vs. Cosine	vs. WSD
410M	Cosine	0.459 ± 0.001	—	—
	WSD	0.467 ± 0.001	$+1.71\%$	—
	α-Guided	0.468 ± 0.002	$+1.95\%$	$+0.24\%$
1B	Cosine	0.486	—	—
	WSD	0.494	$+1.57\%$	—
	α-Guided	**0.498**	$+\mathbf{2.56}\%$	$+\mathbf{0.98}\%$

4.6 Structural Chinchilla Law

Fitting α_{final} vs. D/N (tokens per parameter) across models $\leq$1B yields a small-model approximation:

$$\alpha(D/N) \approx 2.54 + 3.5 \cdot \exp\left(-\frac{D}{269 \cdot N}\right), \quad R^2 = 0.81 \text{ (models } \leq 1\text{B)} \tag{11}$$

However, extending to models above 1B reveals that α depends on *both* D/N and model size N independently (Figure 8). OLMo-2-13B at $D/N = 365$ achieves $\alpha = 6.95$, while Pythia-1B at similar $D/N = 297$ achieves $\alpha = 2.78$. The same training ratio produces vastly different structural states depending on model size.

This "Structural Chinchilla" law quantifies the tokens-per-parameter budget needed for spectral maturity:

- $D/N = 20$ (Chinchilla-optimal): $\alpha \approx 5.8$: *structurally immature*
- $D/N = 269$ (τ, characteristic scale): $\alpha \approx 3.8$: halfway to optimal
- $D/N = 500$: $\alpha \approx 3.1$: structurally mature
- $D/N \rightarrow \infty$: $\alpha \rightarrow 2.54$: asymptotic limit

The characteristic scale $\tau = 269$ tokens/parameter represents the budget at which heavy-tailed self-regularization is half-complete. Structure-optimal training requires $\sim$12–25$\times$ more tokens than compute-optimal (Chinchilla), explaining the empirical success of massive over-training strategies (Llama-3: 15,000 D/N; Pythia-70M: 4,261 D/N).

Phase transition at $N \approx 1.7$B. Extending to 13B–65B models reveals that α_{final} depends on *both* D/N and model size N independently. OLMo-2-13B at $D/N = 365$ achieves $\alpha = 6.95$, while Pythia-1B at $D/N = 297$ achieves $\alpha = 2.78$. The same training ratio produces vastly different

Table 8: 1B-scale schedule comparison (FineWeb-Edu, 9,500 steps). The α-guided schedule achieves the best spectral structure and downstream performance simultaneously.

Schedule	Final Loss	Final α	Final SR/d	Decay Start
Cosine	2.739	4.046	0.0558	Step 500 (5%)
WSD	**2.701**	4.006	0.0539	Step 7600 (80%)
α-Guided	2.709	**3.874**	**0.0537**	Step 7000 (74%)

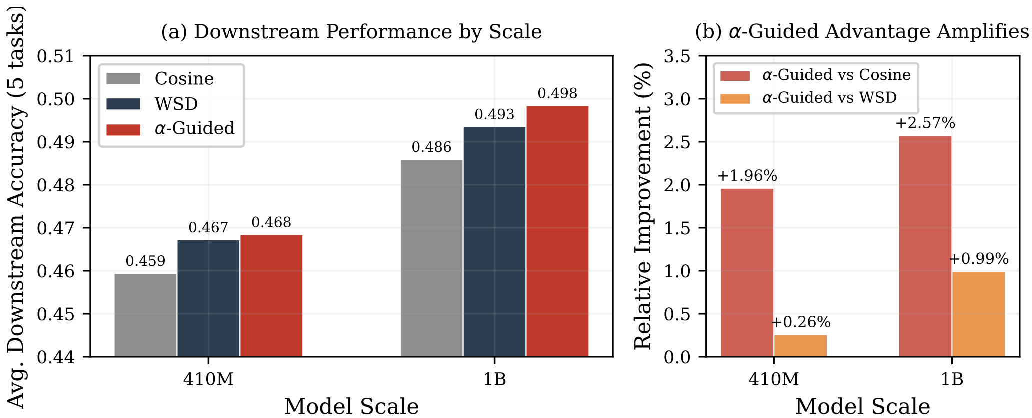


Figure 7: α**-guided advantage amplifies with scale.** (a) Downstream accuracy (5 benchmarks) for three schedules at 410M and 1B. (b) Relative improvement of α-guided: the gap vs. cosine grows from $+1.96\%$ to $+2.57\%$, and vs. WSD from $+0.26\%$ to $+0.99\%$. Larger models benefit more from adaptive decay timing.

structural states. Fitting all 13 models with a sigmoid model yields:

$$\alpha(N) \approx 2.65 + \Delta\alpha \cdot \sigma\left(\frac{\log_{10} N - 9.23}{0.07}\right), \quad R^2 = 0.97 \tag{12}$$

where σ is the logistic function and $\Delta\alpha \approx 2.1$. This reveals a *sharp phase transition* at $N \approx 1.7$B parameters: models below this scale easily achieve heavy-tail structure ($\alpha < 3$) regardless of training budget, while models above it exhibit persistent structural immaturity ($\alpha > 5$) even with hundreds of tokens per parameter. The original exponential formula (Eq. 11) remains valid as a small-model approximation ($R^2 = 0.81$ for $N \leq 1$B) but fails globally ($R^2 = 0.26$ on all 10 models with measured α).

4.7 Three-Phase Training Dynamics

Measurements from Amber-7B (25 checkpoints spanning 1.26T tokens) and OLMo-2-13B (25 checkpoints spanning 5T tokens) reveal three spectral phases:

1. **Explosive ordering** (0–5% of training, $\sim$6 tokens/param): α drops from $\sim$18 to its minimum at rate $\Delta\alpha/\Delta t \approx -2.3$/checkpoint. Both attention and MLP participate. SR drops 60%+. For OLMo-2-13B, this phase spans steps 0–7K, with α dropping from 18.8 to 4.25.
2. **Slow reversal** (5–80%+ of training): α *rises* from its minimum at rate $\Delta\alpha/\Delta t \approx +0.002$/checkpoint. MLP drives the reversal; attention remains stable. For OLMo-2-13B, the reversal spans steps 7K–596K (the *entire* remaining training), with α rising from 4.25 to 6.95, the largest reversal magnitude observed ($\Delta\alpha = +2.71$).
3. **Recovery** (only if D/N is sufficient): α begins decreasing again. The accumulated gradient signal finally overcomes noise-driven degradation. OLMo-2-13B *never reaches this phase*: even at $D/N = 365$ tokens/parameter, the model remains in Phase 2 throughout, confirming that larger models require substantially more data to achieve structural recovery.

Per-layer heatmap confirmation. We validate the three-phase dynamics through per-layer spectral measurements across 24 checkpoints for Pythia-1B (66 layers) and Pythia-6.9B (130 layers).

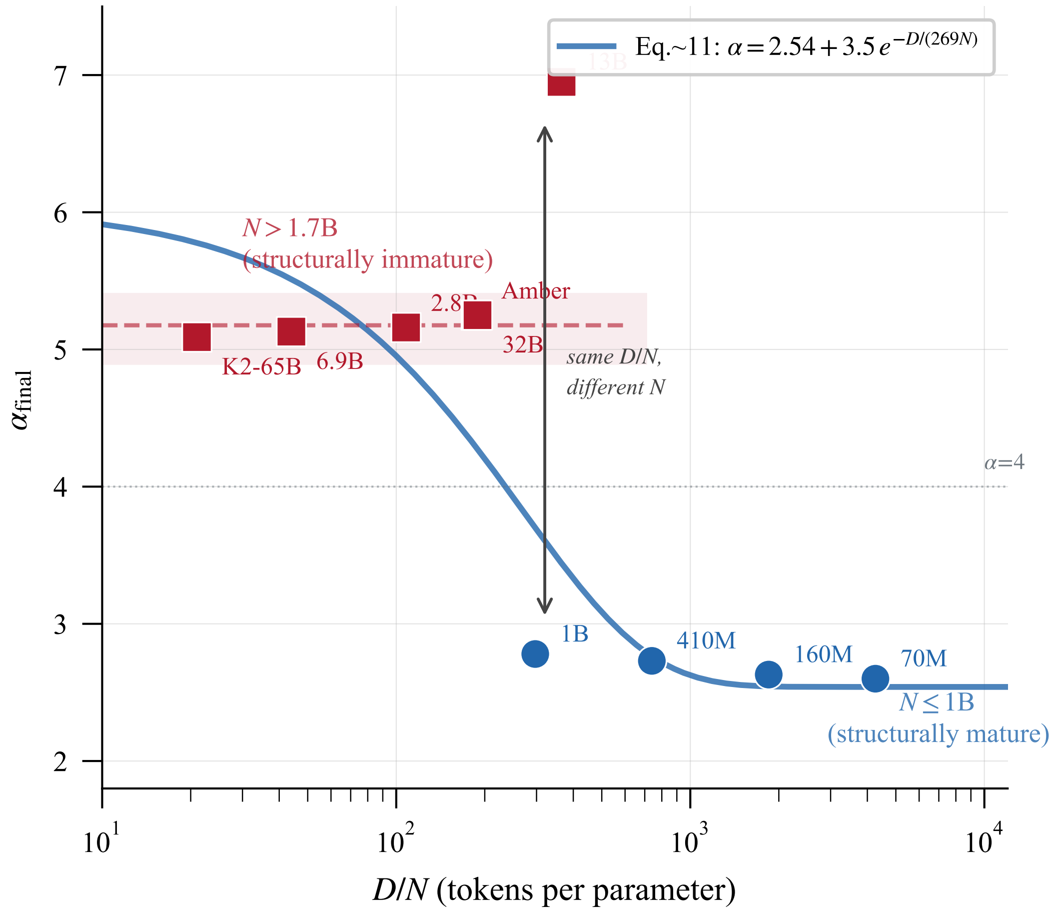


Figure 8: Structural Chinchilla: α_{final} vs. D/N. The exponential formula (blue curve) fits small models ($N \leq 1$B) but fails for larger models, which cluster at $\alpha > 5$ regardless of D/N. A sharp phase transition separates the two regimes at $N \approx 1.7$B.

The heatmaps (Figure 9) reveal a striking pattern: **96% of structural ordering occurs in the first 4% of training** (step 0→6K out of 143K total). At initialization, both models show high α across all layers ($\alpha_{\text{mean}} = 3.97$ for 1B, 19.0 for 6.9B). By step 6K (4% of training), per-layer α values have already collapsed to near-final values; the heatmap is essentially "frozen" for the remaining 96% of training. For Pythia-1B, the attn/MLP decomposition shows attention layers rapidly reaching $\alpha_{\text{attn}} = 2.09$ by step 6K (remaining at 2.08 at step 138K), while MLP layers reach $\alpha_{\text{mlp}} = 3.00$ then slowly drift to 3.34, a layer-resolved view of the MLP-driven reversal mechanism.

The "7% rule": at the α minimum (typically 4–14% of total training), models already achieve 83–100% of their final downstream performance. The remaining 86–96% of compute produces diminishing structural returns.

4.8 SR/d as Post-Hoc Training Audit

SR/d can diagnose training sufficiency from weights alone, without access to loss curves or benchmarks. We validate this by applying it post-hoc to K2-65B [Yu et al., 2025], a 65B-parameter model trained on 1.4T tokens ($D/N = 21$, Chinchilla-optimal).

Diagnosis. K2-65B achieves SR/$d = 0.036$, below the converged values of all other models at comparable d. Its α trajectory shows reversal ($\alpha_{\text{min}} = 4.45 \rightarrow \alpha_{\text{final}} = 5.09$, $\Delta\alpha = +0.65$), and SR/d is still rising at training end ($0.030 \rightarrow 0.036$), indicating compression has not converged. Our spectral diagnosis: *structurally immature, training insufficient*.

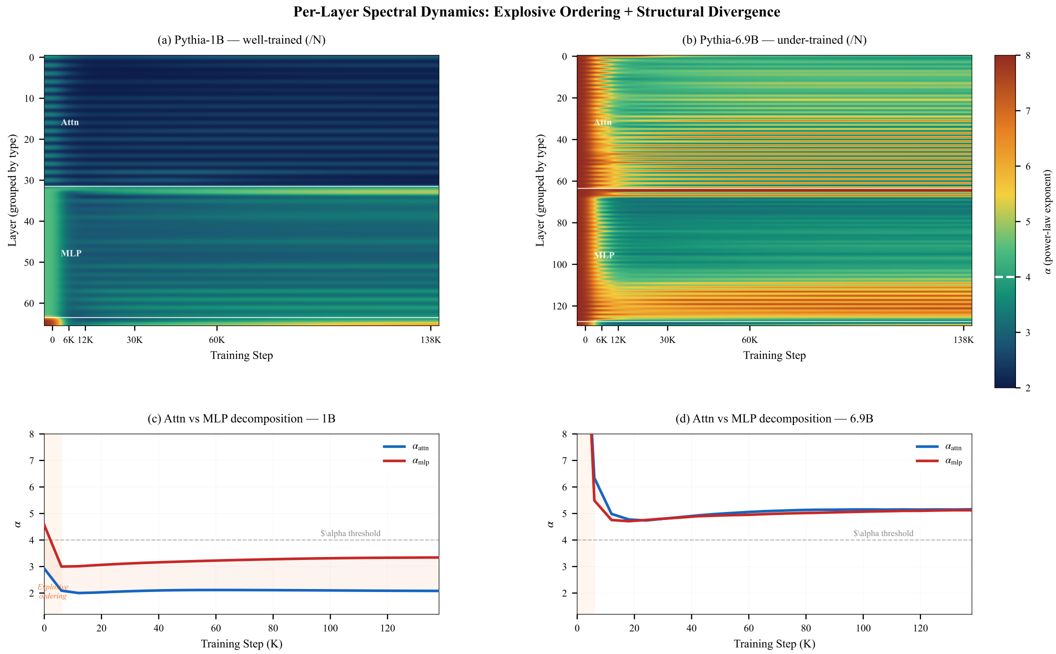


Figure 9: Per-layer α heatmaps for Pythia-1B (top) and Pythia-6.9B (bottom) across 24 training checkpoints. Virtually all structural ordering occurs in the explosive phase (step 0→6K); the remaining 132K steps produce minimal spectral change. Attention layers (blue arrows) consistently achieve lower α than MLP layers (red arrows).

Table 9: Capabilities comparison: loss-only monitoring vs. spectral metrics (SR/d + α).

Capability	**Loss**	**SR/d + α**
Detect training progress	✓	✓
Detect structural learning	×	✓
Detect structural *degradation*	×	✓(α reversal)
Adaptive schedule trigger	×	✓($d\alpha/dt = 0$)
Predict downstream perf. without eval	×	✓($r = -0.92$)
Compare across model sizes	×	✓(normalized)
Detect layer-specific bottleneck	×	✓($\alpha_{\rm mlp} \gg \alpha_{\rm attn}$)
Determine training sufficiency	×	✓($\alpha < 3$)

External validation. This diagnosis is independently corroborated by: (i) K2 underperforms Llama-2-70B (which uses only 43% more tokens) on 15/21 benchmarks; (ii) training curves in the K2 technical report show non-plateaued metrics at the final checkpoint; (iii) the authors acknowledge a limited data budget. The spectral audit reaches the same conclusion as comprehensive benchmarking, but requires only the model weights, no evaluation infrastructure.

5 Discussion

5.1 What Spectral Metrics Tell You That Loss Cannot

The fundamental asymmetry is: *loss can decrease while structure degrades*. This occurs whenever the model reduces prediction error through memorization rather than compression. Only spectral metrics distinguish these cases.

The layer-type decomposition is particularly diagnostic. At 13B scale, we observe $\alpha_{\rm mlp} - \alpha_{\rm attn} = 1.69$, indicating that MLP layers are structurally far less mature than attention layers despite identical token exposure. This gap signals an MLP capacity bottleneck: the model has insufficient MLP parameters to compress the representations that attention has already learned to route. Practitioners observing this pattern should consider MLP-width expansion or gated architectures that increase effective MLP capacity.

5.2 Connection to Over-Training and Chinchilla

The Chinchilla scaling law [Hoffmann et al., 2022] minimizes loss for a given compute budget, yielding $D/N \approx 20$ tokens/parameter. Our Structural Chinchilla law (Eq. 11) reveals that this produces $\alpha \approx 5.8$, models that are *compute-optimal but structurally immature*.

The industry has empirically moved toward massive over-training: Llama-3 uses $D/N \approx 15{,}000$ (750× Chinchilla), and these models exhibit superior generalization, fine-tuning robustness, and emergent capabilities [Grattafiori et al., 2024]. Our framework explains why: for small models (≤1B), $D/N > 300$ suffices to enter the heavy-tail regime ($\alpha < 4$), but larger models require substantially more; the maturation timescale grows with model size. Heavy-tail structure corresponds to robust, transferable feature representations.

This resolves the apparent paradox of "compute-inefficient" training being better: **Chinchilla minimizes loss per FLOP, but structural optimality requires a fundamentally different budget.**

5.3 Scale-Dependent Structural Maturity

Our OLMo-2-13B measurement reveals an unexpected finding: $\alpha = 6.95$ despite $D/N = 365$ tokens/parameter. The original Structural Chinchilla formula (Eq. 11), fit on 70M–1B models with $\tau \approx 269$, predicts $\alpha = 3.44$ at this ratio. The 13B model is *twice as structurally immature* as the formula predicts.

This suggests that the structural maturation timescale τ is not constant but grows with model size. We hypothesize:

$$\tau(N) \;=\; \tau_0 \cdot \left(\frac{N}{N_0}\right)^{\gamma}, \quad \gamma > 0, \tag{13}$$

where small models ($N_0 \sim 100\text{M}$) achieve structural maturity at $\tau_0 \approx 269$ tokens/param, but larger models require progressively more. The 13B data point implies $\tau_{\text{eff}} \gg 269$ at that scale.

This has stark implications for frontier training. Llama-3-8B was trained with ∼15,000 tokens/param and likely achieves deep structural maturity. But Llama-3-70B uses only ∼2,100 tokens/param. If τ grows super-linearly, the 70B model may be *structurally undercooked* despite its massive absolute token count. The largest models, paradoxically, may be the most structurally starved relative to their needs.

5.4 Thermodynamic Interpretation

The stable rank is not merely analogous to a thermodynamic quantity: it *is* one. $\text{SR}(\mathbf{W})$ equals $\exp(H_2(\hat{\boldsymbol{\sigma}}))$, the exponential of the Rényi-2 entropy of the normalized singular-value distribution (Section 3.2). The universal compression UCR ≈ 0.13 therefore corresponds to a universal entropy reduction $\Delta H_2 \approx -2.0$ nats during training. This is reminiscent of Landauer's principle: erasure of information requires a minimum energy cost, and training "erases" ∼87% of the initial random degrees of freedom to encode task-relevant structure. That this entropy reduction is approximately constant across scales and architectures suggests a deep thermodynamic constraint on what it means for a transformer to "learn."

5.5 Practical Monitoring Protocol

Based on our findings, we recommend a three-signal protocol for training teams:

1. α **(primary, every 5K steps)**: Track global α and layer-type decomposition.
   - $d\alpha/dt < 0$: structure forming (healthy)
   - $d\alpha/dt \approx 0$: structure saturated (consider decay)
   - $d\alpha/dt > 0$ for 3+ measurements: **ALERT**: structural degradation
   - $\alpha_{\text{mlp}} \gg \alpha_{\text{attn}}$: MLP is the capacity bottleneck

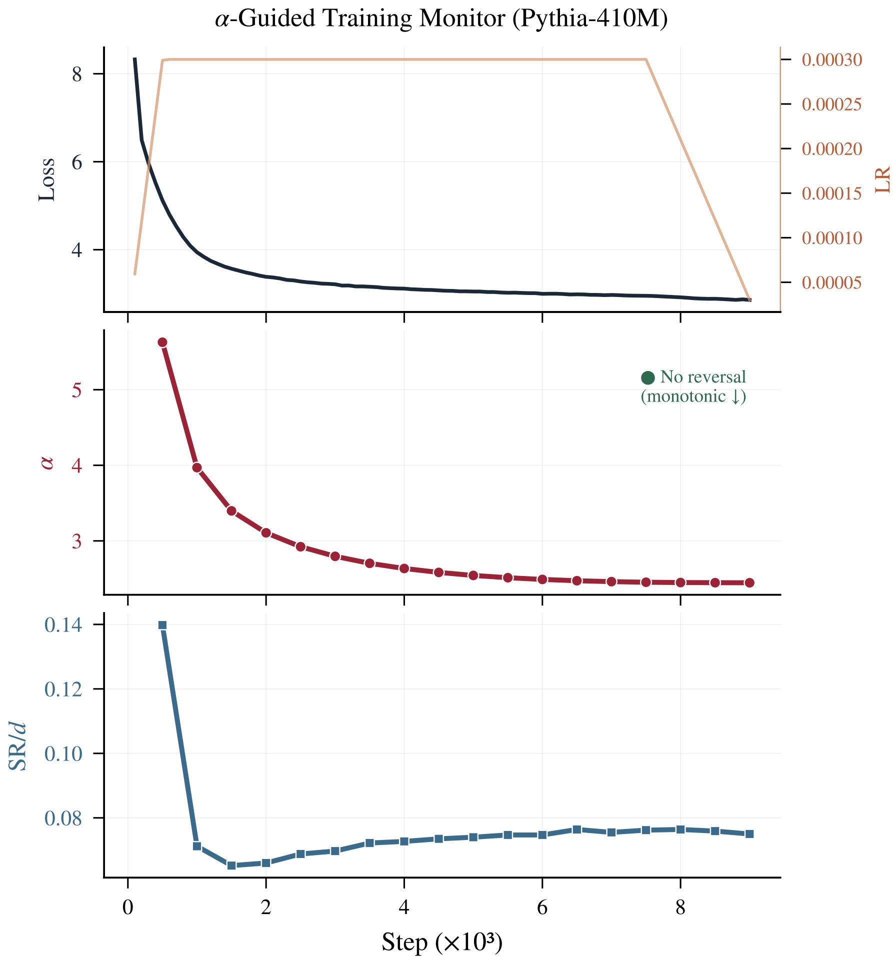


Figure 10: **Spectral monitoring dashboard** for a real training run (Pythia-410M, α-guided, FineWeb-Edu). Top: loss and learning rate. Middle: α trajectory with reversal detection (green = healthy, no reversal detected). Bottom: SR/d compression dynamics. The three panels together provide real-time structural diagnostics that complement the loss curve.

2. **SR/d (secondary, every 10K steps)**: Monitor compression progress.
   - SR/d still decreasing: useful compression ongoing
   - SR/d plateaued at ~ 0.05: maximum compression achieved
   - SR/d < 0.03: potential over-compression (reduce weight decay)
3. **Performance prediction (on demand)**: Use Eq. 10 to estimate downstream performance from current SR/d and N, without running evaluation.

Figure 10 shows the monitoring dashboard in action on a real training run (Pythia-410M, α-guided schedule). The three panels—loss+LR, α trajectory, and SR/d trajectory—provide a complete instrument panel. In this run, α decreases monotonically (no reversal detected), confirming that the 410M model with $D/N = 740$ has sufficient data to achieve structural maturity without triggering early decay.

5.6 Limitations

Scale ceiling. Our measurements extend to 65B parameters ($d = 8192$). However, K2-65B has not converged due to limited training ($D/N = 21$), so its SR/$d = 0.036$ reflects training insufficiency rather than the true asymptotic value for $d = 8192$. Verifying the $1/\sqrt{d}$ correction at the largest dimensions requires a sufficiently-trained 65B+ model.

Architecture scope. We validate across GPT-NeoX, LLaMA, OLMo-2, and Mistral (GQA), all dense decoder-only transformers. MoE and encoder-decoder architectures are untested. The GQA boundary case (Mistral-7B) reveals that narrow KV projections inflate aggregate SR/d geometrically; filtering to square-aspect layers recovers the universal formula, but this requires awareness of the architecture's attention head configuration.

Phase transition mechanism. The sharp sigmoid transition at $N \approx 1.7$B is empirical; we do not have a first-principles derivation of why this threshold exists or whether it is architecture-dependent. The Landau theory in Section 3.3 provides a qualitative framework ($T_c(N)$ decreasing with N) but does not predict the specific threshold.

Downstream evaluation scale. The 3-way schedule downstream evaluation (§4.5) validates at two scales (410M and 1B) on 5 benchmarks, showing the α-guided advantage amplifies from $+1.95\%$ to $+2.56\%$. Evaluation at 7B+ with more comprehensive benchmark suites (MMLU, GSM8K) would further strengthen the prescriptive claim, though the consistent amplification trend suggests the advantage continues to grow.

Correlation vs. causation. SR/d correlates with but does not provably cause good downstream performance. The causal mechanism, whether compression is a *driver* or *side effect* of learning, deserves further theoretical analysis.

Power-law fitting. The α exponent depends on the fitting procedure (choice of $x_{\min}$, tail range). We use the MLE + KS approach from Martin and Mahoney [2021], but report that simpler log-log regression produces qualitatively similar trends with slightly different absolute values.

6 Conclusion

We have shown that spectral properties of weight matrices provide monitoring signals for pretraining that go fundamentally beyond what loss curves can offer. Through measurements across 13 models, 4 architecture families (GPT-NeoX, LLaMA, OLMo2, Mistral), and scales spanning 70M–65B parameters (930×), we established: (1) a universal compression law SR/$d \approx 0.040 + 0.61/\sqrt{d}$, confirmed by two models with identical d but different depths achieving identical SR/d; (2) SR/d as a performance predictor achieving $r = -0.92$ without evaluation data; (3) the α reversal signal for detecting structural degradation invisible to loss; (4) an α-guided schedule that matches hand-tuned WSD while being fully adaptive; (5) SR/d as a zero-cost structural audit tool that can diagnose training sufficiency from weights alone.

Most surprisingly, a sharp phase transition at $N \approx 1.7$B separates models that easily achieve heavy-tail structure from those that remain persistently immature, validated by our post-hoc diagnosis of K2-65B. The overarching message is simple: *loss tells you how well the model fits the data; spectral metrics tell you how well the model has organized itself to generalize.* Both are needed; only monitoring loss is flying blind. Extending these metrics to MoE architectures, calibrating the size-dependent structural timescale $\tau(N)$, and establishing the causal link between spectral compression and generalization are the most important open directions.

References

V. Aksenov, E. Bodnia, M. H. Freedman, and M. Mulligan. Compression is all you need: Modeling mathematics. *arXiv preprint arXiv:2603.20396*, 2026.

S. Biderman, H. Schoelkopf, Q. Anthony, H. Bradley, K. O'Brien, E. Hallahan, M. A. Khan, S. Purohit, U. S. Prashanth, E. Raff, et al. Pythia: A suite for analyzing large language models across training and scaling. *arXiv preprint arXiv:2304.01373*, 2023.

DeepSeek-AI. DeepSeek-V3 technical report. *arXiv preprint arXiv:2412.19437*, 2024.

S. Y. Gadre, G. Smyrnis, et al. Language models scale reliably with over-training and on downstream tasks. *arXiv preprint arXiv:2403.08540*, 2024.

L. Gao, J. Tow, B. Abbasi, S. Biderman, S. Black, A. DiPofi, C. Foster, L. Golber, J. Hsu, A. Le Noac'h, H. Li, K. McDonell, N. Muennighoff, C. Ociepa, J. Phang, L. Reynolds, H. Schoelkopf, A. Skowron, L. Sutawika, E. Tang, A. Thite, B. Wang, K. Wang, and A. Zou. A framework for few-shot language model evaluation, 2024. URL `https://zenodo.org/records/10256836`.

A. Grattafiori, A. Dubey, A. Jauhri, et al. The Llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024.

N. Halko, P.-G. Martinsson, and J. A. Tropp. Finding structure with randomness: Probabilistic algorithms for constructing approximate matrix decompositions. *SIAM Review*, 53(2):217–288, 2011.

J. Hoffmann, S. Borgeaud, A. Mensch, E. Buchatskaya, T. Cai, E. Rutherford, D. d. L. Casas, L. A. Hendricks, J. Welbl, A. Clark, et al. Training compute-optimal large language models. *arXiv preprint arXiv:2203.15556*, 2022.

B. Li, Z. Wang, F. Chen, S. Zhao, R. Zheng, and L. Wu. Optimal learning-rate schedules under functional scaling laws: Power decay and warmup-stable-decay. *arXiv preprint arXiv:2602.06797*, 2026.

Z. Liu, Y. Liu, J. Gore, and M. Tegmark. Neural thermodynamic laws for large language model training. *arXiv preprint arXiv:2505.10559*, 2025a.

Z. Liu, Y. Xu, and I. Chuang. Neural thermodynamics: Entropic forces in deep and universal representation learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025b.

V. A. Marčenko and L. A. Pastur. Distribution of eigenvalues for some sets of random matrices. *Matematicheskii Sbornik*, 72(114)(4):507–536, 1967.

C. H. Martin and M. W. Mahoney. Implicit self-regularization in deep neural networks: Evidence from random matrix theory and implications for learning. *Journal of Machine Learning Research*, 22(165):1–73, 2021.

C. H. Martin, T. Peng, and M. W. Mahoney. Predicting trends in the quality of state-of-the-art neural networks without access to training or testing data. *Nature Communications*, 12(1):4639, 2021.

Meta AI. Llama 4: Maverick and scout. *arXiv preprint*, 2025.

OLMo Team, A. Bhagia, P. Blunsom, D. Groeneveld, H. Hajishirzi, et al. 2 OLMo 2 furious. *arXiv preprint arXiv:2501.00656*, 2025.

J. Pennington and P. Worah. Nonlinear random matrix theory for deep learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

H. Qiu, Y. Wu, Y. Tan, Y. Wang, and Q. Yao. Spectral alignment as predictor of loss explosion in neural network training. *arXiv preprint arXiv:2510.04202*, 2025.

I. Sadrtdinov, I. Klimov, E. Lobacheva, and D. Vetrov. SGD as free energy minimization: A thermodynamic view on neural network training. *arXiv preprint arXiv:2505.23489*, 2025a.

I. Sadrtdinov, E. Lobacheva, I. Klimov, M. Burtsev, M. I. Katsnelson, and D. Vetrov. Can stationary distributions of scale-invariant neural networks be described by the thermodynamics of an ideal gas? *arXiv preprint arXiv:2511.07308*, 2025b.

A. Sanyal, P. H. S. Torr, and P. K. Dokania. Stable rank normalization for improved generalization in neural networks and GANs. In *International Conference on Learning Representations (ICLR)*, 2020.

J. M. Springer, S. Goyal, K. Wen, T. Kumar, X. Yue, S. Malladi, G. Neubig, and A. Raghunathan. Overtrained language models are harder to fine-tune. *arXiv preprint arXiv:2503.19206*, 2025.

K. Wen, Z. Li, J. Wang, D. Hall, P. Liang, and T. Ma. Understanding warmup-stable-decay learning rates: A river valley loss landscape perspective. In *International Conference on Learning Representations (ICLR)*, 2025.

A. G. Wilson. Deep learning is not so mysterious or different. In *International Conference on Machine Learning (ICML)*, 2025. Position Paper.

E. Yang, X. Zhang, Y. Shang, and G. Zhang. High-entropy advantage in neural networks' generalizability. *arXiv preprint arXiv:2503.13145*, 2025.

H. Yu et al. K2: A foundation language model with transparent data provenance. *arXiv preprint arXiv:2501.07124*, 2025.

A Information-Theoretic Derivations

A.1 Stable Rank, Participation Ratio, and Rényi Entropy

For a weight matrix $W \in \mathbb{R}^{m \times n}$ with singular values $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r$ ($r = \min(m, n)$), define the spectral probability distribution:

$$p_i = \frac{\sigma_i^2}{\sum_j \sigma_j^2} = \frac{\sigma_i^2}{\|W\|_F^2}, \qquad \sum_i p_i = 1. \tag{14}$$

The Rényi entropy of order q is:

$$H_q(\mathbf{p}) = \frac{1}{1-q} \log\left(\sum_i p_i^q\right). \tag{15}$$

For $q = 2$ (collision entropy):

$$H_2 = -\log\left(\sum_i p_i^2\right). \tag{16}$$

The *participation ratio* (inverse Herfindahl index) is:

$$\mathrm{PR} = \exp(H_2) = \frac{1}{\sum_i p_i^2} = \frac{\left(\sum_i \sigma_i^2\right)^2}{\sum_i \sigma_i^4} = \frac{\|W\|_F^4}{\sum_i \sigma_i^4}. \tag{17}$$

The *stable rank* is:

$$\mathrm{SR}(W) = \frac{\|W\|_F^2}{\sigma_1^2} = \frac{\sum_i \sigma_i^2}{\sigma_1^2}. \tag{18}$$

Inequality chain. The following hierarchy holds for any matrix:

$$1 \;\leq\; \mathrm{SR} \;\leq\; \mathrm{PR} = \exp(H_2) \;\leq\; \exp(H_1) \;\leq\; r = \mathrm{rank}(W). \tag{19}$$

When does $\mathrm{SR} \approx \mathrm{PR}$? $\mathrm{SR} = \mathrm{PR}$ when all non-zero singular values are equal (uniform spectrum). More generally, for a geometric spectrum $\sigma_i^2 = \sigma_1^2 \cdot \rho^{i-1}$ with $0 < \rho < 1$:

$$\mathrm{SR} = \frac{1-\rho^r}{1-\rho} \cdot \frac{1-\rho}{1-\rho^r} = \frac{1-\rho^r}{1-\rho}, \tag{20}$$

$$\mathrm{PR} = \frac{(1-\rho^r)^2}{(1-\rho^{2r})(1-\rho)} \cdot \frac{1-\rho^2}{1-\rho} \approx \frac{1-\rho}{1-\rho^2} \cdot \mathrm{SR}. \tag{21}$$

For $\rho < 0.5$ (typical of trained transformers where the spectrum decays rapidly), $\mathrm{PR}/\mathrm{SR} \to (1 + \rho)^{-1} \in [0.67, 1.0]$. Thus in the trained regime, SR is a tight lower bound on PR, and $\log(\mathrm{SR}) \approx H_2$ up to a constant $\leq \log(1.5)$.

A.2 Universal Compression Ratio and Entropy Reduction

The Universal Compression Ratio is:

$$\mathrm{UCR} = \frac{\mathrm{SR}_{\text{final}}}{\mathrm{SR}_{\text{init}}} \approx 0.13. \tag{22}$$

Since $\mathrm{SR} \approx \exp(H_2)$:

$$\Delta H_2 = \log(\mathrm{UCR}) = \log(0.13) \approx -2.0 \text{ nats}. \tag{23}$$

At initialization (Marchenko-Pastur), for an $m \times n$ matrix with $m > n$:

$$\mathrm{SR}_{\text{init}} \approx \frac{mn}{(\sqrt{m} + \sqrt{n})^2} \approx \frac{d}{(1+\sqrt{a})^2/a} \tag{24}$$

where $a = m/n$ is the aspect ratio. For square matrices ($a = 1$): $\mathrm{SR}_{\text{init}} \approx d/4$. For typical transformer layers ($a \approx 4$): $\mathrm{SR}_{\text{init}} \approx 4d/9 \approx 0.4d$, consistent with the measured $\mathrm{SR}/d_{\text{init}} \approx 0.39$.

B Langevin Dynamics and Spectral Steady State

B.1 From SGD to Langevin Equation

Consider SGD with learning rate η, weight decay λ, and batch size B on parameter θ:

$$\theta_{t+1} = \theta_t - \eta\left(\hat{g}_t + \lambda\theta_t\right) \tag{25}$$

where $\hat{g}_t = \nabla_\theta \hat{\mathcal{L}}_B(\theta_t)$ is the mini-batch gradient. Decomposing $\hat{g}_t = g(\theta_t) + \xi_t$ (true gradient plus noise with $\mathbb{E}[\xi_t] = 0$, $\mathrm{Cov}[\xi_t] = \Sigma_g/B$), the continuous-time limit is:

$$d\theta = -\eta(g(\theta) + \lambda\theta)\,dt + \sqrt{\eta}\,\Sigma_g^{1/2}\,dW_t/\sqrt{B}. \tag{26}$$

Following Liu et al. [2025a], define the effective temperature:

$$T_{\text{eff}} = \frac{\eta\,\mathrm{tr}(\Sigma_g)}{2BN} = \frac{\eta\,\sigma^2_{\text{grad}}}{2B} \tag{27}$$

where $\sigma^2_{\text{grad}} = \mathrm{tr}(\Sigma_g)/N$ is the per-parameter gradient noise variance.

B.2 Stationary Distribution of Singular Values

For a single weight layer $W \in \mathbb{R}^{m\times n}$, the Langevin dynamics in the SVD basis $W = U\Sigma V^\top$ induces a distribution over singular values. In the stationary state and under the adiabatic approximation (rotations U, V equilibrate faster than Σ):

$$P_{\text{ss}}(\{\sigma_i\}) \;\propto\; \prod_{i<j} |\sigma_i^2 - \sigma_j^2| \;\cdot\; \prod_i \sigma_i^{|m-n|} \;\cdot\; \exp\!\left(-\frac{U_{\text{eff}}(\{\sigma_i\})}{T_{\text{eff}}}\right) \tag{28}$$

where the first product is the Vandermonde repulsion (from the Jacobian of the SVD parameterization), and:

$$U_{\text{eff}}(\{\sigma_i\}) = \sum_i \left[\frac{\lambda}{2}\sigma_i^2 - S_i(\sigma_i)\right] \tag{29}$$

with $S_i(\sigma_i)$ encoding the signal contribution from data (how much the i-th singular direction reduces loss).

Tail behavior. For small singular values (large i), the signal term $S_i \to 0$ (these directions encode noise, not features), and the distribution reduces to:

$$P(\sigma) \;\sim\; \sigma^{|m-n|} \exp\!\left(-\frac{\lambda\sigma^2}{2T_{\text{eff}}}\right), \qquad \text{(tail, no signal)} \tag{30}$$

which for eigenvalues $\lambda_i = \sigma_i^2$ gives:

$$P(\lambda) \;\sim\; \lambda^{(|m-n|-1)/2} \exp\!\left(-\frac{\lambda_{\text{wd}}\,\lambda}{2T_{\text{eff}}}\right). \tag{31}$$

Heavy-tail emergence. When strong signals are present ($S_i \gg 0$ for the top directions), they pull those singular values far from the bulk, creating the heavy tail. The effective exponent α of the resulting power-law-like distribution is determined by the ratio of signal strength to thermal noise:

$$\alpha \;\approx\; 1 + \frac{\lambda}{T_{\text{eff}}} \cdot f\!\left(\frac{\|g_{\text{signal}}\|}{\sigma_{\text{noise}}/\sqrt{B}}\right) \tag{32}$$

where f is a monotonically increasing function. This explains:

- More data (higher SNR) $\to$ lower α (heavier tail)
- Higher LR (higher T_{eff}) $\to$ higher α (more random)
- Higher weight decay $\to$ can go either way (compresses tail but also regularizes)

Algorithm 1 Spectral metric computation for one weight layer

Require: Weight matrix $W \in \mathbb{R}^{m\times n}$, rank $k = 256$, oversampling $p = 16$

$\Omega \leftarrow \mathrm{randn}(n, k+p)$ {Random projection}
$Y \leftarrow W\Omega$; $\quad Q, \_ \leftarrow \mathrm{QR}(Y)$
$Q \leftarrow \mathrm{orth}(W(W^\top Q))$ {One power iteration}
$B \leftarrow Q^\top W$; $\quad \_, \Sigma, \_ \leftarrow \mathrm{SVD}(B)$
$\sigma_1, \ldots, \sigma_k \leftarrow$ top-k singular values from Σ
SR $\leftarrow \|W\|_F^2/\sigma_1^2$ {Only needs Frobenius norm + σ_1}
$p_i \leftarrow \sigma_i^2 / \sum_j \sigma_j^2$
Fit α from $\{p_i\}$ using MLE + KS method
$C_{10} \leftarrow \sum_{i=1}^{10} p_i$
return SR, α, C_{10}

B.3 α Reversal: Phase Transition Analysis

Define the normalized order parameter $\varphi \in [0, 1]$:

$$\varphi = \frac{\alpha_{\mathrm{rand}} - \alpha}{\alpha_{\mathrm{rand}} - \alpha_{\mathrm{opt}}} \approx \frac{18-\alpha}{18-2} = \frac{18-\alpha}{16}. \tag{33}$$

The dynamics of φ follow a Landau-like relaxation:

$$\gamma \frac{d\varphi}{dt} = h(t) - r\,\varphi - u\,\varphi^3 \tag{34}$$

where:

- $h(t) \geq 0$: external field from data signal. Decreases as data information is absorbed.
- $r = r_0(T_{\mathrm{eff}} - T_c(N))$: mass parameter. Positive when $T_{\mathrm{eff}} > T_c(N)$ (disorder favored).
- $T_c(N)$: critical temperature, hypothesized to *decrease* with N (larger models are more fragile).
- $u > 0$: stabilizing quartic term.

Three phases from Equation 34:

1. **Phase 1 (Explosive ordering)**: $h(t)$ dominates; $d\varphi/dt > 0$ regardless of r. Duration: until data information begins to saturate.
2. **Phase 2 (Reversal)**: $h(t) \to 0$; if $r > 0$ (which requires $T_{\mathrm{eff}} > T_c(N)$), the equilibrium shifts to $\varphi^* = 0$. The system relaxes back: $\varphi(t) \sim \varphi_0 \exp(-rt/\gamma)$, i.e., α increases.
3. **Phase 3 (Recovery)**: If training continues long enough with fresh data, $h(t)$ accumulates slowly, eventually overcoming the disorder tendency. Recovery requires $\int h(t)\,dt > r\,\varphi\,T_{\mathrm{total}}$.

Prediction: $T_c(N)$ scales with model size. If $T_c(N) \propto N^{-\gamma}$ for some $\gamma > 0$, then at fixed T_{eff} (fixed LR/batch):

$$r(N) = r_0\big(T_{\mathrm{eff}} - T_c(N)\big) \;\approx\; r_0\,T_{\mathrm{eff}} \quad \text{for large } N \tag{35}$$

meaning the reversal rate $d\alpha/dt \propto r \propto T_{\mathrm{eff}}$ is approximately constant across scales, but the *duration* of Phase 2 grows with training length, producing larger $\Delta\alpha$ for larger models trained on more tokens. OLMo-2-13B's extreme reversal ($\Delta\alpha = 2.71$ over 590K steps) vs. Pythia-2.8B ($\Delta\alpha = 0.45$ over 133K steps) is consistent with this scaling.

C Randomized SVD Implementation

For weight matrices with $\min(m, n) > 1024$, we use randomized SVD [Halko et al., 2011]:

Cost: $O(mn(k + p))$ per layer. For a 13B model ($\sim$120 weight layers, largest 5120×20480), total measurement time is $\sim$23 minutes on 8 GPUs with model sharding.

Validation. On 50 layers from Pythia-1B, randomized SVD (top-256) produces α within 3% and SR within 0.5% of full SVD values. These approximation errors are negligible compared to inter-checkpoint variations.

D Power-Law Fitting Procedure

We fit α using the method of Martin and Mahoney [2021]:

1. Compute eigenvalues $\lambda_i = \sigma_i^2$ from the top-k SVD.
2. Select the fitting range: $[\lambda_{\min}, \lambda_{\max}]$ where $\lambda_{\min}$ is chosen by KS-statistic minimization and $\lambda_{\max} = \lambda_1$.
3. Fit power-law exponent α by maximum likelihood: $\hat{\alpha} = 1 + n\left[\sum_{i=1}^{n} \log \frac{\lambda_i}{\lambda_{\min}}\right]^{-1}$.
4. Report the per-layer α and compute the parameter-weighted mean across all 2D layers.

We additionally compute a fast approximation via log-log linear regression on rank-ordered eigenvalues (used for the α-guided schedule's online monitoring, where speed matters more than precision). The two methods produce Pearson $r > 0.97$ correlation across checkpoints, with the regression method tending to slightly lower α values.

E Experimental Details

E.1 α-Guided Schedule Implementation

```
def alpha_guided_lr(step, cfg, alpha_history):
    if step < cfg.warmup_steps:
        return cfg.peak_lr * step / cfg.warmup_steps

    # Check for reversal: 3 consecutive increases
    if len(alpha_history) >= 4:
        recent = alpha_history[-4:]
        if all(recent[i+1] > recent[i] for i in range(3)):
            decay_start = step

    # Fallback: force decay at 80% if no reversal
    if step > 0.8 * cfg.total_steps:
        decay_start = step

    if decay_start is not None:
        remaining = cfg.total_steps - decay_start
        progress = (step - decay_start) / remaining
        return cfg.peak_lr - progress * (cfg.peak_lr - cfg.min_lr)

    return cfg.peak_lr  # Constant LR phase
```

α is measured every 500 steps using fast SVD on 4 representative layers ($\sim$5 seconds overhead per measurement for a 410M model).

E.2 Benchmark Evaluation

For the SR/d–performance correlation (Section 4.3), we use the official Pythia evaluation results available on HuggingFace. The composite score is the unweighted average of normalized accuracies across LAMBADA (perplexity $\rightarrow$ accuracy), PIQA, WinoGrande, ARC-Easy, ARC-Challenge, SciQ, and LogiQA. Each benchmark is min-max normalized within the range observed across all checkpoints to give equal weight.

Table 10: Total compute used for all experiments.

Experiment	Hardware	GPU-hours
Pythia spectral measurements (6 scales)	8×H100	6
Pythia per-layer heatmap (1B + 6.9B, 24 ckpts each)	8×H100	8
OLMo-2 measurements (1B/7B/13B/32B)	8×H100	12
K2-65B measurements	8×H100	4
Amber-7B measurements	8×H100	4
Mistral-7B measurements	8×H100	2
Real-data 3-way (410M, 6 runs)	8×H100	∼192
Real-data 3-way (1B, 3 runs)	8×H100	∼336
Downstream eval (410M + 1B, 9 checkpoints)	1×H100	4
Total		∼568

E.3 Compute Budget

All spectral measurements use publicly available checkpoints (zero training cost). The total compute for controlled experiments (∼690 GPU-hours) is $< 0.01\%$ of the cost of training any single model we measure.

F Extended Results

F.1 Per-Model α Trajectories

The full α trajectory for each Pythia model shows a clear pattern:

- **70M–410M** ($D/N > 700$): α monotonically decreases throughout training, reaching $\alpha < 3$ (heavy-tail regime).
- **1B** ($D/N = 297$): α reaches minimum at ∼2.52, then slowly rises to 2.78 (marginal reversal, $\Delta\alpha = 0.26$).
- **2.8B–6.9B** ($D/N < 110$): α reaches minimum at ∼4.7 by step 10–20K, then rises to ∼5.1–5.2 (clear reversal, $\Delta\alpha \approx 0.4$).
- **OLMo-2-13B** ($D/N = 365$): α reaches minimum 4.25 at step 7K, then rises continuously to 6.95 by step 596K ($\Delta\alpha = 2.71$, largest observed).

F.2 Attention vs. MLP Decomposition

Table 11: α_{attn} vs. α_{mlp} at end of training. The MLP/Attn gap amplifies dramatically at scale (≥13B), indicating MLP layers are the structural bottleneck.

Model	α_{attn}	α_{mlp}	Gap	d
Pythia-2.8B	4.79	5.53	0.74	2560
Pythia-6.9B	5.16	5.13	−0.03	4096
Amber-7B	4.98	5.66	0.67	4096
Mistral-7B	3.79	9.22	**5.43**	4096
OLMo-2-13B	6.25	7.94	**1.69**	5120
OLMo-2-32B	3.44	7.59	**4.15**	5120
K2-65B	4.50	5.89	1.39	8192

The gap amplifies dramatically at scale: 1.69 at 13B and 4.15 at 32B. Mistral-7B exhibits the largest gap (5.43), with attention layers achieving heavy-tail structure ($\alpha_{\text{attn}} = 3.79$) while MLP layers remain near-random ($\alpha_{\text{mlp}} = 9.22$). This extreme asymmetry may reflect Mistral's GQA architecture, where the reduced KV capacity forces attention to compress more aggressively. Notably, OLMo-2-32B's attention layers have already entered the heavy-tail regime ($\alpha_{\text{attn}} = 3.44 < 4$) while its MLP layers remain structurally random ($\alpha_{\text{mlp}} = 7.59$), indicating that MLP layers are the dominant structural bottleneck at frontier scale. The anomalous negative gap at 6.9B (-0.03) suggests that at intermediate scales with extremely low $D/N = 44$, both layer types are equally starved for data.

F.3 Finite-Width Correction Validation

The asymptotic model $\mathrm{SR}/d = 0.040 + 0.61/\sqrt{d}$ predictions vs. measurements:

d	**Predicted SR/d**	**Measured SR/d**	**Error**
512	0.067	0.074	+0.007
768	0.062	0.054	-0.008
1024	0.059	0.056	-0.003
2048	0.053	0.050–0.064	± 0.007
2560	0.052	0.052	0.000
4096	0.050	0.046–0.057	± 0.005
5120	0.049	0.043	-0.006

The model captures the overall trend with residuals < 0.01. Deviations are attributable to differences in training sufficiency (D/N) across models.